The Riemann Hypothesis via Band-Limitedness

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Abstract

Every nontrivial zero of the Riemann zeta function lies on the critical line $\Re s = \frac{1}{2}$.

1 Axioms

Axiom 1 (Hardy Z-function). *$Z(t) := e^{i\vartheta(t)}\zeta(\frac{1}{2}+it)$ with $\vartheta(t) := \arg\Gamma(\frac{1}{4}+it/2) - (t/2)\log\pi$. Z is real on $\mathbb{R}$, and for $t \in \mathbb{R}$, $Z(t) = 0 \iff \zeta(\frac{1}{2}+it) = 0$.*

Axiom 2 (Digamma asymptotic). *$\vartheta'(t) = \frac{1}{2}\log|t/(2\pi)| + O(|t|^{-2})$, $\vartheta''(t) = (2t)^{-1} + O(t^{-3})$, and $c_\star := -\inf_{\mathbb{R}}\vartheta' \in (0,\infty)$.*

Axiom 3 (Riemann–Siegel). *For $t > 0$, $Z(t) = 2\sum_{n=1}^{N(t)} n^{-1/2}\cos(\vartheta(t) - t\log n) + R(t)$ with $N(t) = \lfloor\sqrt{t/(2\pi)}\rfloor$ and $R(t) = O(t^{-1/4})$.*

Axiom 4 (Band-limited Paley–Wiener). *Let $f \in L^2_{\mathrm{loc}}(\mathbb{R})$ and $K_T(\mu) := (2\pi)^{-1}\int_0^T f(u)e^{-i\mu u}\,du$. If $|K_T(\mu)|^2 \le MT$ uniformly in μ, T for some $M > 0$, and $\lim_{T\to\infty}|K_T(\mu)|^2/T = 0$ for every $|\mu| > 1$, then f extends to an entire function on $\mathbb{C}$ of exponential type ≤ 1 with Fourier transform supported in $[-1,1]$ as a tempered distribution.*

Axiom 5 (Akhiezer–Laguerre–Pólya). *If $F : \mathbb{C} \to \mathbb{C}$ is entire of exponential type ≤ 1, real on $\mathbb{R}$, with distributional Fourier transform supported in $[-1,1]$, then every zero of F in $\mathbb{C}$ lies in $\mathbb{R}$.*

2 Construction

Definition 1. $\Theta(t) := \vartheta(t) + ct$, any fixed $c > c_\star$.

Definition 2. $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$ for $u \in \mathbb{R}$.

Definition 3. $K[T](\mu) := (2\pi)^{-1}\int_0^{\Theta(T)} Y(u)e^{-i\mu u}\,du$.

Lemma 4 (Warp). *$\Theta : \mathbb{R} \to \mathbb{R}$ is a C^∞ bijection with $\inf_{\mathbb{R}}\Theta' = c - c_\star > 0$; in particular $Z(t) = \sqrt{\Theta'(t)}\,Y(\Theta(t))$ with $\sqrt{\Theta'(t)} > 0$ for every $t \in \mathbb{R}$.*

Proof. $\Theta' = \vartheta' + c \ge c - c_\star > 0$ by Axiom 2, so Θ is a strict C^∞ bijection. The factorization is the definition of Y inverted. ☐

Lemma 5 (Frequency ratio). *$\omega_n(t) := (\vartheta'(t) - \log n)/\Theta'(t)$ satisfies $\omega_n(t) \in [0,1)$ for $t \ge 2\pi n^2$, with $\omega_n(t) \uparrow 1$ as $t \to \infty$.*

Proof. $\vartheta'(t) \ge \log n$ iff $\log(t/(2\pi)) \ge 2\log n$ up to $O(t^{-2})$, i.e. $t \ge 2\pi n^2$ (Axiom 2). Then $0 \le \vartheta' - \log n < \vartheta' + c = \Theta'$, so $\omega_n \in [0,1)$; $\vartheta'(t) \to \infty$ forces $\omega_n \to 1$. ☐

3 Proof of RH

Theorem 6 (Uniform L^2 bound)**.** *There is $M > 0$ with $|K[T](\mu)|^2 \le M\,\Theta(T)$ for every $T \ge 2$ and every $\mu \in \mathbb{R}$.*

Proof. Cauchy–Schwarz: $|K[T](\mu)|^2 \le (2\pi)^{-2}\Theta(T)\int_0^{\Theta(T)} Y^2\,du = (2\pi)^{-2}\Theta(T)\int_0^T Z^2\,dt$ by $u = \Theta(t)$. The Riemann–Siegel diagonal $\int_0^T Z^2 = O(T\log T) = O(\Theta(T))$ (Axioms 2, 3). $\square$

Theorem 7 (Band-limit)**.** *For $|\mu| > 1$, $\lim_{T\to\infty} |K[T](\mu)|^2/\Theta(T) = 0$.*

Proof. Substitute Axiom 3 into $K[T](\mu)$ and change variables $u = \Theta(t)$. Mode (n,σ), $\sigma \in \{\pm 1\}$, $n \ge 1$:

$$\mathcal{J}_{n,\sigma}(T,\mu) = \int_{2\pi n^2}^{T} n^{-1/2}\sqrt{\Theta'(t)}\,e^{i\Phi(t)}\,dt, \qquad \Phi'(t) = \Theta'(t)(\sigma\omega_n(t) - \mu).$$

By Lemmas 4, 5: $|\sigma\omega_n(t) - \mu| \ge |\mu| - 1 =: \lambda > 0$ and $\Theta'(t) \ge c - c_\star$, hence $|\Phi'(t)| \ge \lambda(c - c_\star) > 0$ uniformly. One integration by parts:

$$|\mathcal{J}_{n,\sigma}(T,\mu)| \le \frac{2\,n^{-1/2}\sqrt{\Theta'(T)}}{\lambda(c - c_\star)} + \int_{2\pi n^2}^{T}\left|\frac{d}{dt}\frac{n^{-1/2}\sqrt{\Theta'(t)}}{\Phi'(t)}\right| dt.$$

The derivative is $O(n^{-1/2}\Theta'(t)^{-1/2}\vartheta''(t))$; with $\Theta' \asymp \log t$ and $\vartheta'' \asymp 1/t$ (Axiom 2), the integral is $O(n^{-1/2})$. Hence $|\mathcal{J}_{n,\sigma}(T,\mu)| = O(n^{-1/2}\sqrt{\log T})$.

The remainder $\int_0^T R(t)\sqrt{\Theta'(t)}e^{-i\mu\Theta(t)}dt$ has phase slope $|\mu\Theta'(t)| \ge |\mu|(c - c_\star)$; one IBP gives $O(T^{-1/4})$.

Sum over $1 \le n \le N(T) = O(T^{1/2})$ and $\sigma = \pm 1$: $|K[T](\mu)| = O(T^{1/4}\sqrt{\log T})$, so $|K[T](\mu)|^2/\Theta(T) = O((\log T)/\sqrt{T}) \to 0$. $\square$

Theorem 8 (Entirety and band)**.** *Y extends to an entire function on $\mathbb{C}$ of exponential type ≤ 1 with distributional Fourier transform supported in $[-1,1]$.*

Proof. Theorems 6, 7 supply the two hypotheses of Axiom 4 with $\tau = 1$, applied to $f = Y$ on $[0,\Theta(T)]$. Y is real on $\mathbb{R}$: Lemma 4 gives $Y(\Theta(t)) = Z(t)/\sqrt{\Theta'(t)}$, and Z is real by Axiom 1; since Θ^{-1} is a real bijection, Y is real-valued on $\mathbb{R}$. $\square$

Theorem 9 (Riemann Hypothesis)**.** *Every nontrivial zero of ζ has $\Re s = \frac{1}{2}$.*

Proof. By Theorem 8, Y is entire of exponential type ≤ 1, real on $\mathbb{R}$, with distributional Fourier transform supported in $[-1,1]$. Axiom 5 gives every zero of Y in $\mathbb{C}$ is real. By Lemma 4, $Z(t) = \sqrt{\Theta'(t)}\,Y(\Theta(t))$ with $\sqrt{\Theta'(t)} > 0$, so the zero set of Z on $\mathbb{R}$ is Θ^{-1} of the zero set of $Y|_{\mathbb{R}}$. Y is entire of exponential type and not identically zero ($Y(\Theta(t)) = Z(t)/\sqrt{\Theta'(t)}$ and $Z \not\equiv 0$), so its zeros form a discrete subset of $\mathbb{C}$; all of them lie in $\mathbb{R}$. Axiom 1: $\zeta(\frac{1}{2} + it) = 0 \iff Z(t) = 0 \iff Y(\Theta(t)) = 0$ for $t \in \mathbb{R}$, and a nontrivial zero ρ of ζ off the critical line would give a non-real t with $Z(t) = 0$, hence a non-real zero of Y, contradicting the reality just established. $\square$